

86 Billion Capacitors: The Physics of a Thought

From a dead frog's leg (1791) to brain-computer interfaces (2024).



Membrane =
Capacitor



Ion gradients =
Battery



Ion channels =
Switch



Axon =
Leaky transmission line



Myelin =
Insulation



Across every neuron, the membrane field is about 10^7 V/m.

The Frog, the Fight and the Battery

1791 – 1800



1791 • Galvani

Frog legs twitch. Galvani argues animals make their own electricity.



1800 • Volta

Volta tries to disprove Galvani — and invents the first battery.



The brain is electrochemical.

Chemistry charges the battery. Physics carries the signal.



“Artificial electric organ” — Volta, describing the battery.

Part 1: The Capacitor

The membrane



Capacitance:

$$C = \epsilon A / d$$



Resting voltage:

$\approx -70 \text{ mV}$
across 5 nm.

$\sim 5 \text{ nm}$
(thickness)

Outside the cell
(extracellular fluid)

Lipid bilayer
(dielectric, ϵ)

Inside the cell
(intracellular fluid)



Measured membrane
capacitance:

$\approx 0.9\text{--}1.0 \mu\text{F}/\text{cm}^2$



Electric field:

$E \approx 1.4 \times 10^7 \text{ V/m}$



About 5 × the field
that makes air spark
($3 \times 10^6 \text{ V/m}$).



$Q = CV \rightarrow$ about **6,000 ions per μm^2** move in a spike, only a tiny fraction of the ions inside the cell.

Part 2: The Battery

Ion gradients

Inside vs. Outside

Concentrations in a typical neuron (millimolar, mM).

Ion	Inside (mM)	Outside (mM)
K ⁺	~140	~5
Na ⁺	~10–15	~145

Nernst equation

$$V = \frac{RT}{zF} \ln \left(\frac{C_{\text{out}}}{C_{\text{in}}} \right)$$

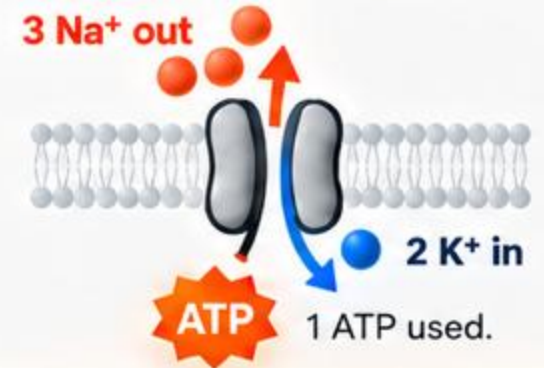
At 37 °C, each tenfold ratio gives about 61.5 mV.

$$E_K \approx -90 \text{ mV}$$

$$E_{\text{Na}} \approx +60 \text{ to } +70 \text{ mV}$$

Sodium–potassium pump

Maintains the gradients using ATP.



Evolution solved the circuit problem too.

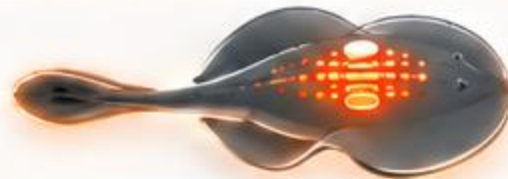
Nature stacks or parallels cells to get the voltage and power it needs.



Electric eel

~6,000 cells in series

→ **860 V**



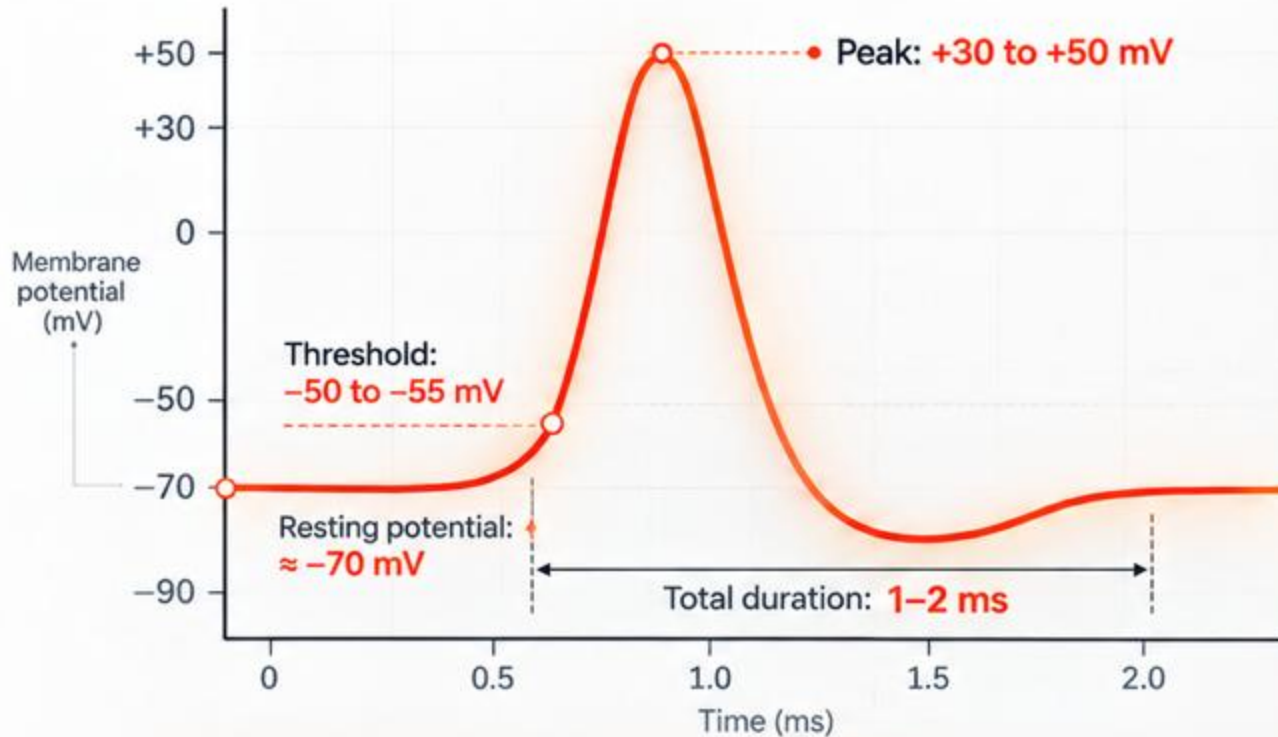
Torpedo ray

cells in parallel

→ **~50 V**,
but over **1 kW** peak pulse

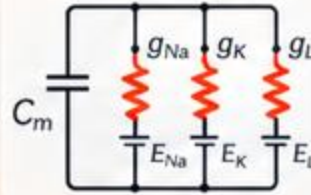
Part 3: The Switch

The action potential



All-or-none spike

Once threshold is crossed, the spike fires completely.



Hodgkin-Huxley circuit

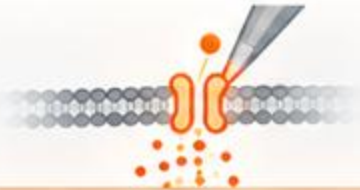
$$C_m \frac{dV}{dt} = -g_{Na}(V - E_{Na}) - g_K(V - E_K) - g_L(V - E_L) + I$$



Proof it worked

Predicted speed: **18.8 m/s**

Measured in squid axon: **21.2 m/s**



Patch clamp measures current through a single channel: about **1-2 pA** ($\sim 10^7$ ions/s).

Part 4: The Wire

The transmission line

Copper wire

signal speed
 $\sim 2 \times 10^8$ m/s



1850 • Helmholtz
measures nerve
speed ≈ 27 m/s.



Cable equation:
signal delay grows
with length².



Myelin = insulation
→ lower capacitance,
higher resistance.



Saltatory conduction:
spike jumps
node to node.

Axon

leaky RC cable



Signals spread and leak
instead of racing like in metal.

Body nerves: up to 1–1.5 mm apart • Brain: 30–150 μ m



Unmyelinated:
0.5–2 m/s



Thick myelinated fibres:
80–120 m/s



Clinical application:

Multiple sclerosis strips myelin,
so conduction slows or fails.

Part 5: The Code

And the one chemical step



Neurons code by **frequency**

Neurons mainly encode information by how often they fire, not by the size of each spike.

Low firing rate



High firing rate



Same spike size, different frequency.



Signal strength is coded by how often the neuron fires.



Refractory period $\approx 1 \text{ ms}$ $\rightarrow$ theoretical ceiling near 1 kHz.



Most neurons stay at a few hundred Hz.



This is closer to FM than AM.



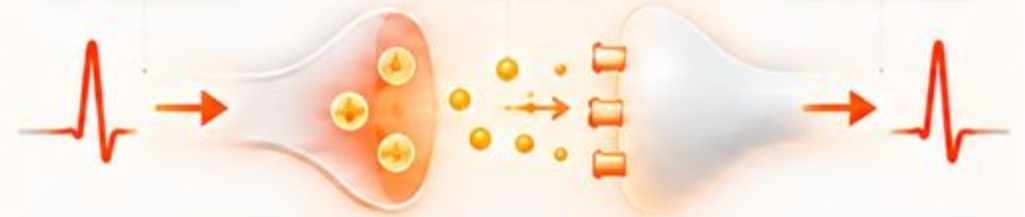
Synapse: **electrical** $\rightarrow$ **chemical** $\rightarrow$ **electrical**

A neuron's electrical signal is converted to a chemical signal at the synapse, then back to an electrical signal in the next neuron.

Electrical signal in

Chemical signal across

Electrical signal out



Typical synaptic delay:
0.5–1 ms



Minimum delay:
about 0.3 ms



Electrical synapses (gap junctions) also exist and pass current almost instantly.

The Scale

How big is the system?



86 billion
neurons



About
150 trillion
synapses in the
neocortex alone.



About
69 billion
are in the cerebellum ($\approx 80\%$),
even though it is only
 $\sim 10\%$ of brain mass.



**150,000 –
180,000 km**
of myelinated wiring at age 20 –
about **4 times around Earth**.



The cortex has about
16 billion
neurons.



Power: about
20 W



An NVIDIA H100 draws
700 W
– about **35× more**.

Reading the Capacitors

Each tool is a different chapter of physics



EEG

10–100 μV on the scalp;
alpha waves 8–13 Hz.



Physics:
Electrostatics



fMRI

Proton spin, $\gamma/2\pi = 42.58 \text{ MHz/T}$
→ about 128 MHz at 3 T.



Indirect measures blood
oxygen, not neurons.



Physics:
Quantum spin and magnetism



MEG

10 fT–1 pT, about 10^8 – 10^9
times weaker than Earth's field.



Physics:
Superconductivity (SQUIDs)



PET

Positron + electron →
two 511 keV gamma rays.



Physics:
Antimatter and $E = mc^2$



Writing Back

From reading the brain to changing it



TMS

A magnetic pulse (~ 1.5 T in ~ 200 μ s) induces currents and makes neurons fire.



FDA cleared in 2008 for depression that has not responded to medication.



Focused ultrasound

- 1,024 emitters focus on a millimetre-scale target through the skull.
- MRI monitors temperature in real time.



FDA 2016 for tremor.



Cochlear implants

Electrical pulses directly stimulate the auditory nerve.



About 1 million implants by 2022.



Brain-computer interfaces

2006

BrainGate cursor control

2023

speech decoding at 62 and 78 words/min

Jan 2024

first Neuralink human patient



**A thought is charge moving across 86 billion capacitors —
and physics has learned to read it.**